



AutoValuate Intelligence

Know what your car is really worth, with the reasoning shown, not hidden.

Computer vision + explainable ML + agentic RAG, live in production for the UAE used-car market.

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A working product, deployed and free

- Enter a car, optionally add photos. You get a priced range with the drivers shown, a damage read, market context and a written, citation-checked report.
- Fifteen result panels cover the full decision: value it, judge the deal, test assumptions, finance it, repair it, time the sale, negotiate it, list it.
- Runs entirely on free-tier infrastructure. No sign-up, no paid tiers.

Deployment status (verified today)

Frontend on Vercel, API on Render, data on Supabase. Live at auto-valuate-intelligence.vercel.app. Weekly data refresh and CI gates run on GitHub Actions.

13.18%

median pricing error, held-out folds

AED 0

cost to the user

15

result panels in the live app

142

tests in the committed suite

The seller has a feeling. The dealer has the data.

- Used-car pricing in the UAE is opaque. Sellers negotiate on a hunch against people who price cars all day.
- Existing checkers return one number with no reasoning, no damage awareness and no honesty about uncertainty.
- So the number can't be defended in a negotiation, which is where it's actually needed.

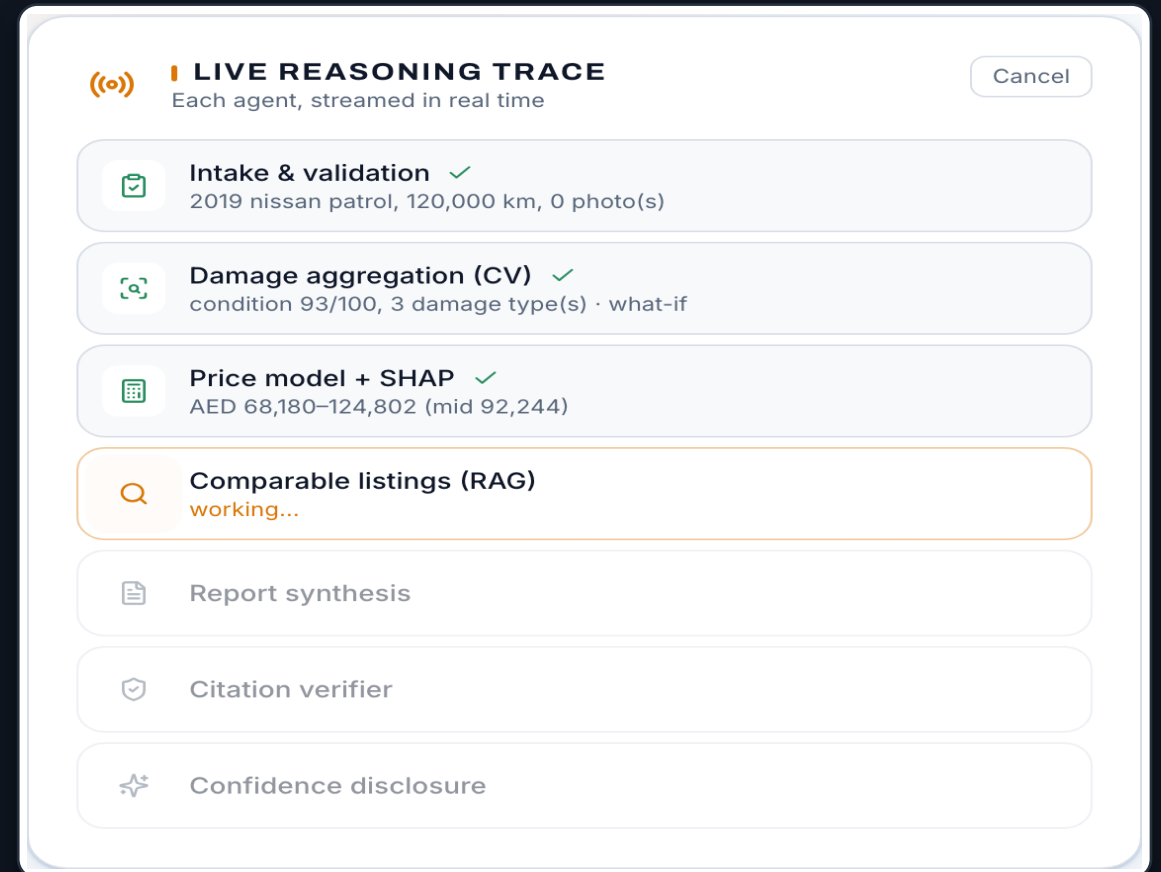
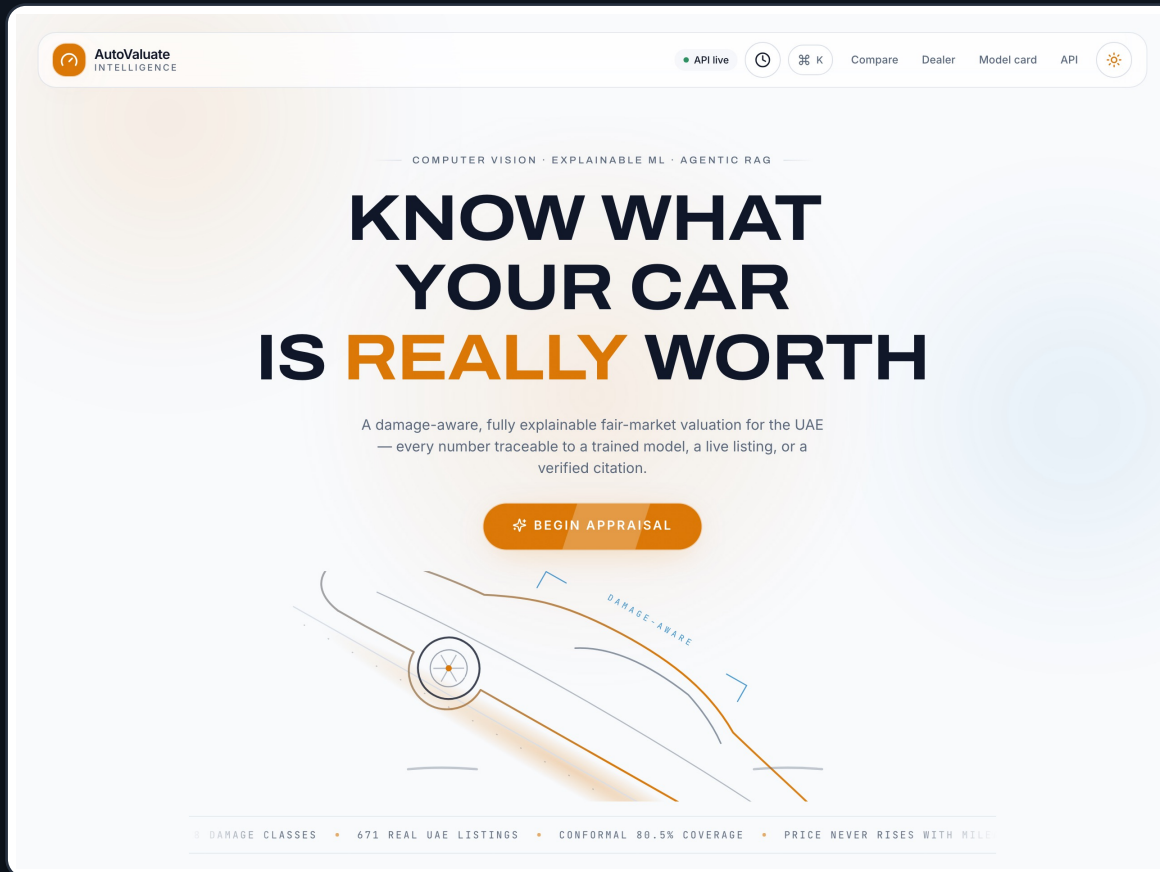
The gap

Nobody gives a private seller an explainable, damage-aware, uncertainty-honest price they can argue with. That is the product.

Who it serves

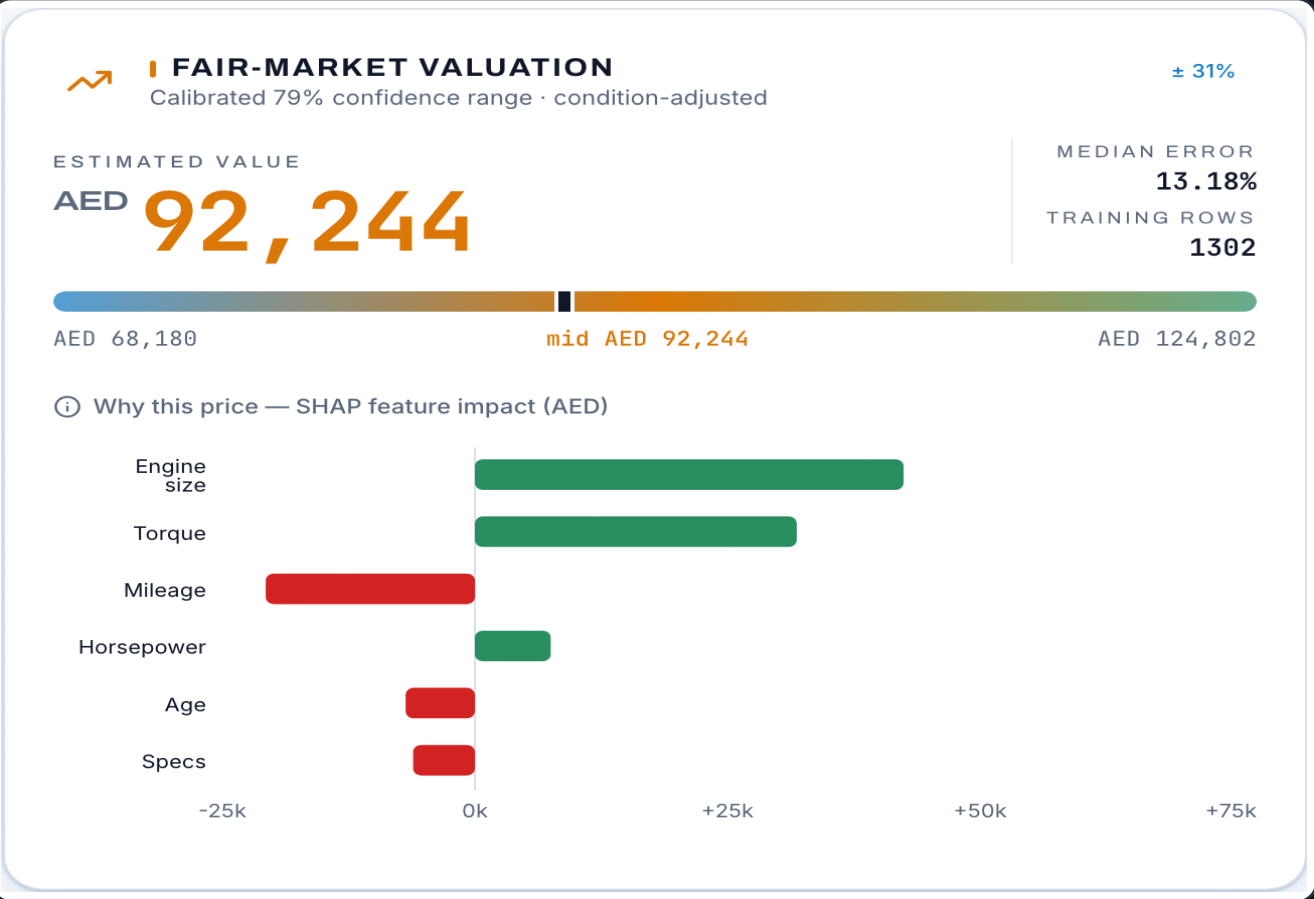
Private sellers pricing a car before listing it.
Buyers checking whether an ask is fair before viewing.
Small dealers valuing a fleet by CSV without a data team.
Developers, through an open API with no key.

From car details to a defensible answer



Details in (photos optional) → each agent streams its step live → valuation, decision tools, report and share.

An explainable price, with honest uncertainty



13.18%

median APE, 5 seeds x 5 folds, held out vs a make and model median baseline

+52.3%

79%

conformal interval coverage, target 80% of cars matched to physical specs

81%

How

XGBoost on 1,302 real UAE listings, joined to vehicle specs (power, torque, weight). SHAP shows each factor's AED effect. Split-conformal bands, calibrated per tier.

Damage scanning on your device, coached at capture

VISUAL DAMAGE ASSESSMENT

3 photo(s) scanned · on-device · tap a finding to price a repair

on-device

-7% value

93

/ 100

Dent

×3

74%

-AED 3,989

Scratch

×2

58%

-AED 1,496

Lamp Broken

×1

81%

-AED 1,994

Tap a detected issue to see what fixing it recovers.

CONDITION-ADJUSTED
AED 92,743

Condition breakdown — value impact by damage type

Damage Type	Count	Percentage	Value Impact
Dent	3	74%	-AED 3,989
Scratch	2	58%	-AED 1,496
Lamp Broken	1	81%	-AED 1,994

Walk around your car

1 of 8 angles captured.

13%

Front

Front right

Right side

Rear right

Rear

Rear left

Left side

Front left

Front: Too far away — step closer so the car fills more of the frame.

Retake

New: capture coaching

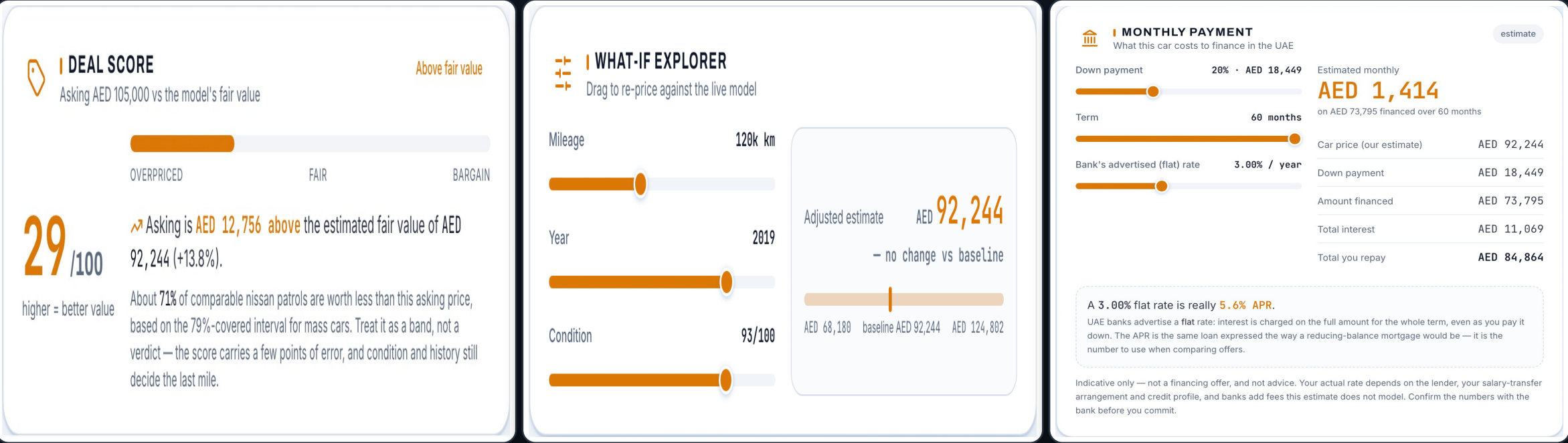
A 3.5 MB Apache-2.0 detector (YOLOX-Nano) checks each shot as it's taken: too far, cut off, blurry, too dark. Advice only. It never touches the score.

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From fair value to a real purchase decision

AutoValuate now helps users judge the deal, test assumptions and understand financing affordability.



Repair, depreciation and when to sell

 **REPAIR ESTIMATE**

What the detected damage would cost to put right

3 items

Estimated repair cost

AED 1,290 – AED 5,520

Dent ×3 moderate

AED 900 – AED 3,600

Lamp Broken minor

AED 210 – AED 1,200

Scratch ×2 minor

AED 180 – AED 720

 This damage is costing you about AED 7,479 in value, against a repair bill of roughly AED 3,405. Repairing before you sell is likely to pay for itself.

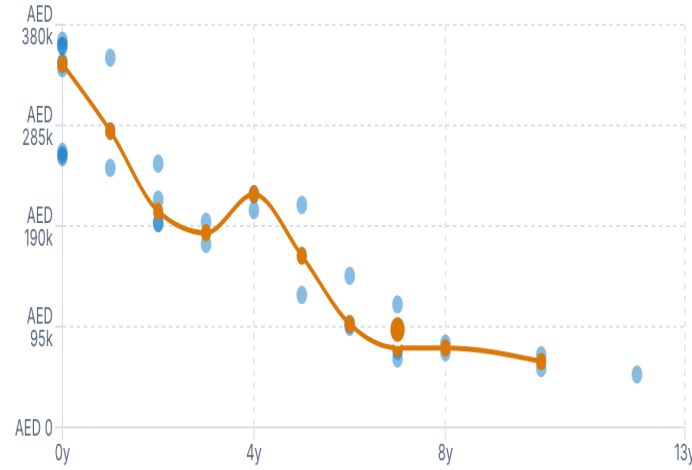
Indicative UAE independent-workshop prices from a published table, scaled by detected severity and instance count — not a quote. Get a workshop to confirm before you decide.

 **DEPRECIATION CURVE**

What the market asks for a Nissan Patrol at every age

38 live listings

Price vs age — each dot is a live listing, the line is the median over all 38, your car in amber



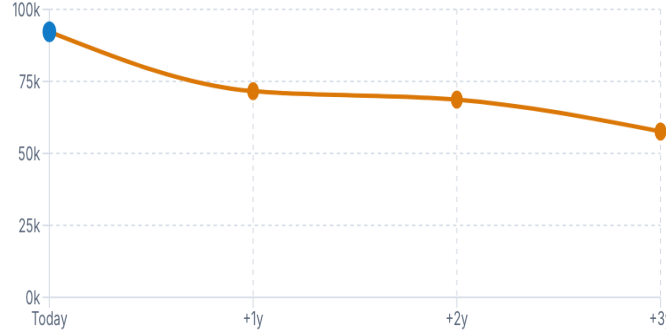
Age (y)	Median Price (AED)
0	350,000
1	280,000
2	220,000
3	190,000
4	200,000
5	170,000
6	130,000
7	110,000
8	100,000
9	95,000
10	90,000
11	85,000
12	80,000
13	75,000

These are asking prices of live UAE listings, not sale prices — sellers typically settle below them. Your car sits at the model's mid estimate; ages with a single listing don't get a median point.

 **SELL-TIMING FORECAST**

What the pricing model says this car will be worth as it ages

-22% / year

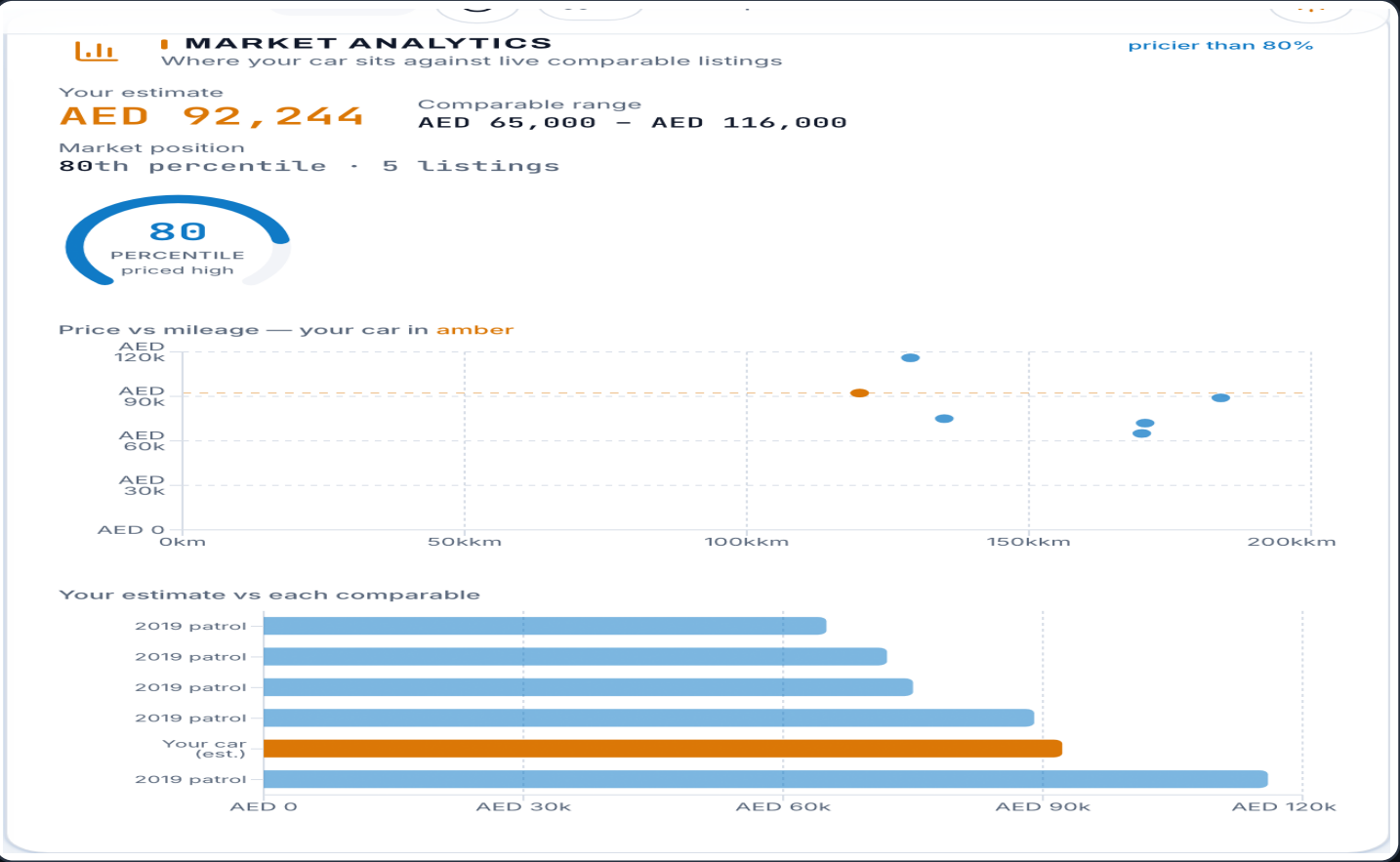


Time	Projected Value (AED)
Today	90,000
+1y	70,000
+2y	68,000
+3y	55,000

Holding this car costs roughly AED 1,722 a month in lost value (about AED 20,666 over the next year). It's depreciating quickly — selling sooner preserves more of its value.

Projected by re-running the pricing model with this car aged forward (age +N years, mileage +N × its own annual rate) — not a generic depreciation rate. It carries the same 13.18% median error as any other estimate, and it assumes the market itself stays flat.

Priced against a live market, not a static table



1,307

P@5 1.0

live UAE listings in the corpus

same-make retrieval precision

Kept honest

The corpus refreshes weekly through a scheduled scrape with a hard budget cap.

When a model is too rare to chart, the analytics card says the corpus is thin instead of drawing a misleading graph. Growing data is the fix, and the card says that too.

The report, the negotiation and the listing

SELLER REPORT

Synthesized via groq · every figure citation-grounded

Certificate

PDF

Share

verified

The estimated value of this vehicle falls within a range of AED 68,180 **v1** to AED 124,802 **v3** , with a mid-point of AED 92,244 **v2** . This valuation is driven by various factors, including the vehicle's condition, mileage, and features such as engine size, with an impact of AED 42,438 **p1** , and torque, with an impact of AED 31,862 **p2** .

The condition of the vehicle is relatively good, with a condition score of 93 **d0** . However, there are some damages reported, including 3 instances of dent with a confidence level of 0.74 **d1** , 2 instances of scratch with a confidence level of 0.58 **d2** , and 1 instance of a broken lamp with a confidence level of 0.81 **d3** . These damages have a negative impact on the vehicle's value, with the dent having a 4.0% value impact **d1** , the scratch having a 1.5% value impact **d2** , and the broken lamp having a 2.0% value impact **d3** .

A visual inspection of the vehicle was not available, and therefore, the confidence in the valuation is limited. The model median error percentage is 13.18% **v5** , which indicates that there is some uncertainty in the valuation. To get a more accurate assessment of the vehicle's value, it is recommended to conduct a professional inspection.

Comparing this vehicle to similar listings, we can see that there are other 2019 Nissan Patrol models available in the market, such as listing 16770453 with a price of AED 116,000 **c1** , listing 16908628 with a price of AED 89,000 **c2** , listing 16878285 with a price of AED 75,000 **c3** , listing 16865150 with a price of AED 65,000 **c4** , and listing 16902721 with a price of AED 72,000 **c5** . These listings have different mileage and features, but they can provide a useful reference point for determining the value of this vehicle.

14 numbers and 18 citations checked by the Verifier — all trace to a computed value.

NEGOTIATION COACH

Grounded talking points — every figure traces to the model or a live listing

I'm Selling

I'm Buying

OPEN YOUR ASK AT

AED 124,802

FAIR-MARKET TARGET

AED 92,244

HOLD ABOVE

AED 68,180

- The fair-market mid is AED 92,244. Anchor near the top of the range, AED 124,802, and don't drop below AED 68,180 without a reason. **v2** **v3** **v1**
- Your estimate sits above 80% of 5 comparables — justify it with condition, history, or extras, or expect pushback.
- Cite the top comparable — a 2019 Nissan Patrol at AED 116,000 — to defend a higher ask.
- Buyers will spot the Dent (~AED 7,479 off value). Disclose it up front and pre-price it in — surprises kill deals and trust.
- Model confidence is high (±31% band). The number is tight — negotiate close to the mid.

deterministic · no AI guesswork

Copy script

Send

LISTING PACK

A ready-to-paste ask and listing text — every number is the model's

deterministic · no AI guesswork

RECOMMENDED LISTING PRICE

AED 124,500

Anchored at the top of the fair range so there's room to settle near the AED 92,244 mid — the same corridor the negotiation coach uses.

2019 Nissan Patrol · 120,000 km · Other specs — AED 124,500

title

2019 Nissan Patrol

Mileage: 120,000 km

Body: SUV

Transmission: Automatic

Fuel: Petrol

Cylinders: 8

Specs: Other

Location: Abu Dhabi

Condition: 93/100 on an AI photo scan — disclosed up front: dent (moderate), scratch (minor), broken lamp (minor). Priced accordingly.

Built only from your form entries, the scan, and the model — nothing is embellished. Damage found by the scan is disclosed on purpose: surprises kill deals.

Copy listing

Send

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End-to-end architecture on free tiers

Browser

Next.js UI. Damage scan + capture coach run here in ONNX, photos never leave the device.

Vercel

Hosts the frontend. Deploys from main on every push.

FastAPI on Render

LangGraph pipeline: intake, aggregate, value, retrieve, report, verify, disclose.

Models

XGBoost bundle with baked-in spec table. YOLOv8 damage ONNX. YOLOX coach ONNX.

Supabase

Comparables with pgvector retrieval, public share links.

GitHub Actions

CI gates, weekly corpus scrape, keep-alive crons.

Provenance everywhere: model hashes, preprocessing versions and config versions travel with every scan, and the backend rejects mismatches.

Tested like a product, not a notebook

- 142 committed tests: 72 backend, 70 frontend, run in CI on every push.
- Browser and backend score identically on 56 of 56 pinned fixtures, so the user's score can't drift from the server's.
- Same photo, same score, proven: the scan is bit-deterministic after we replaced GPU canvas resampling with pure JS.
- Every feature shipped behind a measured study. The spec join carried a permutation control. Rejected ideas are documented so nobody retries them.

Retraining, reproducible

Training reads the committed corpus, so a fresh clone rebuilds the shipped artefact. Metrics JSONs regenerate with it, and the model card page renders those files directly. No number is typed by hand.

Safeguards in production

Per-IP rate limiting on an open API. Photo size caps. Provenance rejection of forged scans. A verifier gate in front of every report. Honest fallbacks when the API is cold.

Published like a scorecard, failures included



1.000

report faithfulness; corrupted control scores 0.00

1.3pp

mean gap, promised vs delivered coverage

What we say out loud

The 0.732 detector mAP came from a validation subset covering 6 of 8 classes. We publish it with that caveat, not as a headline.

The detector is framing-sensitive: a small crop can move the condition score by tens of points. Diagnosed, documented, and mitigated by capture coaching until a retrain on real UAE photos.

Pricing floor is data-bound: our learning curve says the current corpus supports ~13%, not the ~8% larger markets reach.

Product surfaces and the path from here



← Back

Fleet valuation

Upload a CSV of your inventory and value every car at once. Runs the same pricing model as a single appraisal.

 YOUR INVENTORY

CSV with a make,model,year,kilometers header — or one description per line

3 vehicles

Upload CSV

Load a sample fleet

Value all 3

Export CSV

Fleet value (mid)

Valued

AED 393,998

3 of 3

Vehicle	Mileage	Low	Mid	High
2019 Toyota Land Cruiser	90,000 km	AED 93,759	AED 126,851	AED 171,624
2021 Nissan Patrol	45,000 km	AED 166,071	AED 224,687	AED 303,990
2018 Honda Civic	120,000 km	AED 31,383	AED 42,460	AED 57,447

Bulk valuations run the pricing model only (no photo scan, no written report), so they are fast — open a single appraisal for the full explainable breakdown of any vehicle.

Four surfaces today

Compare up to four cars as a buyer. Bulk fleet valuation by CSV for dealers. An open developer API, no key needed. The public model card you just saw.

Licensing, stated plainly

The damage model derives from AGPL-3.0 YOLOv8, so closed-source resale needs an enterprise licence or a detector swap. Every new component since is deliberately Apache-2.0.

Next

Retrain the detector on labelled UAE whole-car photos. Grow the corpus past 5,000 listings. Arabic interface. Real pilot users.

CLOSE

An honest number you can argue with.

Live product. Real models. Every claim traceable to a committed artefact.



Try it now

`auto-valuate-intelligence.vercel.app`



Read the code

`github.com/krish2105/AutoValuate-Intelligence`

Product status and features verified against repository commit 14a8408, 27 July 2026.